

CreditBridge Ghana

AI Credit Intelligence Platform

Unlocking Credit for Ghana's Invisible Workforce

AI Solutions Challenge 2026 — Industry 4: Finance & SMEs
Codetrain Africa · AI Talent Accelerator 2026

~85%	44%	GH₵518B	9%
of Ghana's workforce in informal sector	financially excluded despite MoMo access	MoMo value transacted in Dec 2025 alone	bank loans go to small businesses

1. Purpose of This Document

This document presents a new opportunity not included in the Codetrain Africa AI Solutions Challenge Opportunity Bank. It is grounded in original research, current data, and a deep understanding of how Ghana's financial and informal sectors actually work. It follows the same rigorous structure as the 30 curated opportunities to allow direct comparison and evaluation.

The opportunity is positioned within **Industry 4 — Finance & SMEs** and addresses the single most urgent, widest-felt, and investor-ready financial problem in Ghana today: the complete exclusion of informal workers from formal credit — not because they are poor, but because no institution has ever recorded that they can repay.

2. The Problem

Millions of hardworking Ghanaians are locked out of loans — not because they cannot pay back, but because no one has ever tracked that they can.

2.1 The Reality on the Ground

Ghana's informal economy employs approximately **85% of the national labour force** and accounts for between 29% and 40% of GDP, depending on the measurement methodology used. This includes market traders, kayayei (market porters), artisans, trotro drivers, mobile food vendors, tailors, hair stylists, mechanics, and the vast majority of small shop owners across every region.

These workers earn real money. They transact daily. Many have operated their businesses for years. Yet when they walk into a bank or microfinance institution to request a loan — even a modest GH₵500 to restock goods — the answer is almost always no. The reason is not their character or their cash flow. It is that they cannot produce what banks require: a payslip, a formal bank statement, a registered business, or a physical asset to use as collateral.

2.2 What the Data Shows

120%	15–18 mo	44%	17%
Collateral required of loan value by banks	Typical bank loan approval timeline	Ghanaians financially excluded (2024)	Ghana credit-to-GDP (far below peers)

In February 2026, the Ghana National Chamber of Commerce and Industry publicly challenged banks at a joint forum, noting that members still face collateral requirements of up to 120% of loan values, approval processes lasting 15 to 18 months, and outright rejection rates that leave viable businesses without financing — even as the Bank of Ghana's policy rate had been cut significantly.

Ghana's credit-to-GDP ratio stood at just over 17% in 2023 — significantly lower than the regional average — highlighting the depth of the access problem. Meanwhile, only 9% of banking sector loans reach small businesses, despite those businesses representing the majority of economic activity.

2.3 The Paradox: Data Exists, But is Ignored

Here is the central paradox that makes this problem solvable: **the data already exists.**

As of December 2025, Ghana had 80.5 million registered MoMo accounts and 26.7 million active accounts processing 982 million transactions per month worth GH₵518 billion. Every trader paying a supplier, every driver collecting a fare, every food vendor receiving payment — all of it is recorded.

The Bank of Ghana's own fintech division has recognised this and is actively advocating for MoMo transaction histories to serve as credit and insurance passports for informal workers. The BoG's e-Cedi pilot has already demonstrated that transaction-pattern data improves credit risk assessment accuracy by 28% compared to traditional methods. MTN Ghana's YellowCredit programme has shown that users who receive credit score visibility are 64% more likely to make timely payments and 41% more likely to maintain consistent savings.

The problem is not a lack of data. **The problem is that no system exists to translate that data into a language banks will act on.**

2.4 Who Experiences This Problem

User	Their Daily Reality	Their Loan Barrier
Market trader (Makola, Kejetia, Kaneshie)	Buys and resells goods daily via MoMo. Needs GH₵500–5,000 to restock.	No payslip. No registered business. No collateral.
Trotro / taxi driver	Earns daily fares collected via MoMo or cash. Wants to buy own vehicle.	Irregular income. No formal employer. No bank statement.
Kayayei / porter	Carries goods in markets. Saves small amounts daily via Susu.	Lowest rung of credit access. Fully excluded.
Artisan / seamstress / carpenter	Skilled worker with regular customers. Needs tools or materials.	No invoices, no records, no credit history.
Food vendor / street seller	Daily sales, daily costs. Needs working capital between market days.	Cash-based history is invisible to formal lenders.

User	Their Daily Reality	Their Loan Barrier
Rural / peri-urban small shop owner	Supplies a community. Wants to expand stock or premises.	Nearest bank branch may be far. No digital footprint.

2.5 Why Existing Solutions Fail in Ghana

Traditional banks	Require formal documentation, registered businesses, collateral at 120% of loan value, and approval timelines of 15–18 months. Structurally incompatible with informal workers.
Microfinance institutions	Charge extremely high interest rates (sometimes 40–100% annualised), trap borrowers in debt cycles, and still demand group guarantees or physical collateral that many workers cannot provide.
Mobile lending apps (e.g. Fido, Bloom)	Offer small, short-term emergency loans but use opaque scoring, charge very high fees, and do not build a durable credit profile that unlocks formal bank lending.
Susu groups / ROSCAs	Community-based savings and lending that work informally but are not recognised by formal lenders. Cannot be used to build a verifiable credit history.
Current credit bureaux (XDS, CRB Africa)	Primarily capture formal sector and bank-referred data. The vast informal sector remains invisible to bureau records.

3. The Opportunity

Build an AI system that converts a Ghanaian informal worker's existing digital financial behaviour — MoMo transactions, Susu records, utility payments, airtime top-ups — into a structured, lender-readable credit intelligence profile.

CreditBridge Ghana is not a lending app. It does not lend money. It builds the **trust layer** that currently does not exist between informal workers and formal lenders — translating real financial behaviour into structured evidence that banks and microfinance institutions can act on.

The core workflow is straightforward and designed for Ghana's actual infrastructure and literacy realities:

1	Connect User links MoMo account (MTN, Telecel, AirtelTigo) and optionally utility/airtime history via USSD or app
2	Analyse AI models process transaction patterns: income regularity, spending behaviour, savings consistency, network trust signals
3	Score System generates a structured CreditBridge Profile — not just a number, but an explainable evidence dossier
4	Share User consents to share their profile with a partner lender. Lender receives structured, verified financial evidence
5	Access Lender makes a faster, evidence-based decision. User builds a growing credit history with every repayment

3.1 Ghana-Specific Design Decisions

Every design decision has been made with Ghana's specific realities in mind. Solutions that work in Kenya or Nigeria often do not translate directly, and this opportunity has been stress-tested against Ghana's actual infrastructure, literacy landscape, and cultural context.

USSD + App dual access	Ghana's smartphone penetration is growing but uneven. A USSD channel (*123# style) ensures the product works on any mobile phone, including basic feature phones used by rural traders and older market women.
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Local language support	Core interactions available in Twi, Ga, Ewe, and Hausa in addition to English. Voice-based interaction supported for lower-literacy users, reflecting Ghana's strong oral communication culture.
MoMo-first data source	MTN MoMo, Telecel Cash, and AirtelTigo Money collectively cover Ghana's informal economy. No reliance on formal bank accounts, which most target users do not have.
User-controlled data consent	The Bank of Ghana's own fintech guidance emphasises user privacy. CreditBridge puts data sharing consent entirely in the user's hands — they choose what to share, with whom, and for how long.
Explainable scoring	Low digital literacy means a black-box score number is not enough. The system explains in plain language why a score is what it is and what specific behaviours would improve it.
No smartphone required to start	Onboarding and data collection can begin via USSD. App interface adds richer features but is not a prerequisite, ensuring inclusion of peri-urban and rural users.
Susu group integration	Susu savings are deeply embedded in Ghanaian culture. The system can incorporate Susu contribution records as positive creditworthiness signals — a uniquely Ghanaian data source.

4. AI Opportunities

This problem is not solvable with a simple CRUD application. The AI is not decorative. Each AI component below addresses a specific part of the problem that previously could not be automated and that AI materially improves.

Transaction Pattern Analysis (Time-Series ML)	Raw MoMo transaction data contains irregular, unlabelled cash flows. Machine learning models — specifically time-series classifiers and anomaly detection — identify income regularity, seasonal patterns, business vs. personal spending, and repayment behaviour signals. This replaces the job a bank analyst would do manually with a payslip — but works on informal data.
Alternative Credit Scoring (Ensemble Models)	Combines MoMo patterns, utility payment consistency, airtime top-up behaviour, and Susu records into a composite creditworthiness score. Gradient-boosted ensemble models outperform single models on sparse, heterogeneous data of this type. The BoG e-Cedi pilot confirms a 28% accuracy improvement over traditional methods using this approach.
Natural Language Explanation (LLMs)	A score number alone is not useful for a low-literacy user or a loan officer unfamiliar with AI outputs. An LLM generates plain-language explanations of what the score means, why it is at that level, and what the user can do to improve it — in English, Twi, Ga, Ewe, or Hausa.
Document Intelligence (OCR + NLP)	Where users have physical records — market association receipts, Susu passbooks, utility bills, mobile money printouts — OCR and NLP can extract and structure that information to supplement digital transaction data. Critical for older traders who have paper-based financial histories.
Fraud and Anomaly Detection	Prevents gaming of the system by detecting unusual transaction patterns that suggest circular money movements designed to inflate apparent income. Anomaly detection models identify these patterns before they contaminate credit profiles submitted to lenders.
Voice AI / Speech-to-Text (Multilingual)	For users with low literacy, a voice interface allows interaction in local languages. Speech-to-text with Ghanaian language models enables users to ask questions about their profile, understand their score, and navigate the platform without needing to read or type.
Lender-Facing Dashboard (Retrieval-Augmented Generation)	Loan officers at partner banks and MFIs receive a structured evidence dossier, not just a number. RAG allows the system to surface specific transaction evidence that supports the credit assessment, with citations — giving loan officers something they can document and defend in credit committee meetings.

5. Potential Implementers & Partners

CreditBridge Ghana operates as a B2B platform. It does not compete with lenders — it enables them. The following organisations have direct incentive to adopt or partner with the platform, and all operate within Ghana's existing regulatory and infrastructure landscape.

Category	Organisations	Why They Adopt
Microfinance Institutions (MFIs)	ARB Apex Bank network, Opportunity International Ghana, Advans Ghana, Fund for Women's Livelihood (FWL)	Need better credit data to lend responsibly to informal clients. High NPL ratio.
Rural & Community Banks (RCBs)	144 licensed RCBs across Ghana operating under ARB Apex Bank	Serve exactly the target population. Lack internal data infrastructure. CreditBridge provides credit history.
Fintech Lenders	Fido, Bloom Credit, FairMoney Ghana, PeoplePension Trust	Already innovating in alternative scoring. CreditBridge provides richer MoMo transaction data.
Development Bank Ghana (DBG)	Government-backed development finance institution	DBG's mandate is increasing SME lending from 9% to 15% of bank loans. CreditBridge provides credit history.
MoMo Operators	MTN Ghana, Telecel, AirtelTigo	Data monetisation opportunity. Providing anonymised transaction data (with consent) to lenders.
NGOs & Development Organisations	SNV, UNCDF, Women's World Banking, IFC	Financial inclusion is a core mandate. CreditBridge delivers measurable impact.

5.1 Regulatory Alignment

CreditBridge Ghana operates within and benefits from an actively supportive regulatory environment:

- The Bank of Ghana's fintech division is actively advocating for MoMo transaction histories to serve as credit data — CreditBridge is the implementation of that policy direction.
- The new BoG Microfinance Framework (March 2026) requires all MFIs to transition to a new regulatory structure by December 2026 — this transition creates demand for better credit intelligence tools.
- Ghana's Data Protection Act (Act 843) governs user data consent. CreditBridge is designed with consent-first data sharing, making it compliant by architecture.
- The Ghana Interbank Payment and Settlement Systems (GHIPSS) interoperability framework means MoMo data from different operators can be aggregated with appropriate permissions.

6. What Success Looks Like

Success is not defined by features built. It is defined by measurable change in real people's lives and in lenders' operations. The following outcomes are realistic, measurable, and demonstrable.

Metric	Baseline (Current)	Target (12 months post-launch)
Time for informal worker to prepare credit application	Weeks to months (gathering documents that often don't exist)	Under 48 hours via CreditBridge profile generation
Loan approval rate for informal sector applicants at partner MFIs	Very low — most rejected at documentation stage	Measurable increase in approvals for profiled users vs. non-profiled
Loan officer time spent on informal applicant assessment	2–4 hours manual review per application	Reduced by structured evidence dossier from CreditBridge
Credit profile accuracy vs. traditional methods	Traditional scoring: baseline	28% improvement (aligned with BoG e-Cedi pilot benchmark)
User financial behaviour post-profile visibility	Baseline financial behaviour	64% more likely to make timely payments (aligned with MTN YellowCredit research findings)
Partner lenders onboarded (Demo Day → 12 months)	0	2–3 pilot MFI or RCB partners with active CreditBridge integration
Active user credit profiles generated	0	500–2,000 profiles in pilot region (Greater Accra or Kumasi market cluster)

6.1 Demo Day Demonstration

By Demo Day, the team should be able to demonstrate:

- A working CreditBridge profile generated from real or realistic MoMo transaction data
- A scored credit intelligence output with plain-language explanation (in English and at least one local language)
- A lender-facing evidence dossier that a loan officer could use in a credit committee meeting
- At least one interview or testimonial from a real informal worker who experienced the loan access problem
- At least one conversation with a microfinance institution or RCB confirming the problem and interest in the solution
- A measurable demonstration: e.g. time to produce credit profile vs. traditional documentation gathering

7. Responsible AI Considerations

CreditBridge Ghana operates in a high-stakes domain. A wrong credit decision affects a real person's livelihood. The following responsible AI requirements are non-negotiable and must be addressed in the product architecture.

Consent and data ownership	Users must explicitly consent to each data source connected, understand what will be shared with lenders, and be able to withdraw consent at any time. Consent UI must work in local languages.
Explainability	No black-box outputs. Every score must come with a human-readable explanation. Loan officers must be able to understand and justify any decision made using CreditBridge data.
Bias testing	Training data must be tested for demographic bias — particularly gender bias, since women-owned businesses are already the most financially excluded. Model outputs must be audited across user segments.
Human oversight	CreditBridge provides evidence and recommendations. The final lending decision always rests with the human loan officer. The system must make this boundary clear in its interface.
Accuracy thresholds	Low-confidence profiles must be flagged rather than scored. The system must know what it does not know and communicate uncertainty honestly to both users and lenders.
Data security	MoMo transaction data is sensitive financial information. End-to-end encryption, minimal data retention, and compliance with Ghana's Data Protection Act (Act 843) are required.
Avoiding debt traps	The system must not be used to push credit on vulnerable users. Profile generation should empower users with information, not funnel them into loans they cannot afford.

8. Week 1 Validation Plan

Following the Codetrain AI Solutions Challenge framework, the opportunity is a hypothesis until validated with real users. The following plan ensures Week 1 produces genuine evidence, not assumptions.

8.1 Users to Interview (minimum 10)

- 5 informal traders from Makola, Kantamanto, or a local market in your region
- 2 trotro or taxi drivers who use MoMo for daily collections
- 2 artisans or service workers (seamstress, mechanic, hair stylist)
- 1 Susu group leader or member who manages informal savings

8.2 Decision-Makers to Interview (minimum 2)

- 1 loan officer or branch manager at a microfinance institution or rural community bank
- 1 fintech founder or product manager at a Ghanaian lending app (Fido, Bloom, etc.)

8.3 Key Questions to Validate

For users:

- Have you ever tried to get a loan from a bank or MFI? What happened?
- What documents were you asked for? Which ones did you not have?
- Do you use MoMo regularly? For business payments? For receiving money from customers?
- Would you be willing to share your MoMo history with a lender if it helped you get a loan?
- What would make you trust a company to hold that data?
- How do you currently manage when you need money urgently for your business?

For lenders:

- What percentage of informal sector applicants do you reject at the documentation stage?
- What information would you need to feel confident lending to an informal worker?
- Would you use a third-party credit intelligence profile if it was structured and evidence-based?
- What would you need to see to pilot a new credit data tool?
- What are your current NPL rates for informal sector loans, and what drives defaults?

8.4 What Would Change the Direction

Week 1 research may reveal that:

- Users are unwilling to share MoMo data — pivot to Susu records or utility payment history as primary data source
- Lenders already have internal alternative scoring — focus on the user-facing profile tool rather than lender integration
- A specific user segment (e.g. market women's associations) has structured group data — target cooperative-level credit profiling first

- Regulatory constraints on MoMo data sharing are more complex than anticipated — engage BoG fintech sandbox programme

9. MVP Scope (Six Weeks)

One important workflow, done extremely well. Not a platform. Not a marketplace. Not an app with twenty features. A system that takes a real informal worker's MoMo data and produces a structured credit profile a real lender would find useful.

9.1 In Scope for MVP

- MoMo transaction data ingestion (manual upload of MoMo statement PDF or CSV, or API integration if accessible)
- AI transaction categorisation and pattern analysis (income, expenses, savings behaviour, regularity)
- CreditBridge Profile generation with plain-language explanation
- Basic lender-facing evidence dossier output (PDF or structured dashboard view)
- English + Twi language support for user-facing screens
- Simple consent flow: user understands what data is used and approves sharing

9.2 Out of Scope for MVP

- Full MoMo API integration (requires operator agreements — pursue post-Demo Day)
- USSD channel (valuable but complex — ship web/app first, USSD in v2)
- Live lender credit decision integration (demonstrate with a partner MFI in a manual pilot)
- Full multilingual support beyond English and Twi
- Automated loan disbursement (CreditBridge does not lend money)

9.3 The One Thing That Must Work on Demo Day

A real or realistic MoMo transaction history goes in. A structured, explainable, lender-ready credit intelligence profile comes out. A loan officer watching the demo should say: 'I could use this.'

10. The Startup & Investor Case

CreditBridge Ghana is built as a startup from day one, with Demo Day as the first milestone. The following makes the case for why this is not just a good project — it is a fundable, scalable business.

10.1 Why Investors Care

Signal	Evidence
Massive, proven market	85% of Ghana's workforce in informal sector. 26.7M active MoMo users. GH₵518B transacted in a s
Urgent, daily pain	Market traders, drivers, and artisans experience loan rejection every week. This is not a niche proble
Regulatory tailwind	The Bank of Ghana is actively advocating for MoMo-based credit scoring. CreditBridge is ahead of,
No dominant incumbent	No Ghanaian company currently owns the informal credit intelligence space. The gap is real and unc
B2B revenue model	Subscription or per-profile fees from MFIs and banks. Predictable, scalable revenue that does not de
Regional scalability	Same problem exists in Nigeria, Senegal, Tanzania, Uganda. Ghana is the beachhead for a West A
Impact + return alignment	Financial inclusion is a priority for DFIs, impact funds, and development banks (IFC, UNCDF, GIZ). I
Data moat	Every user profile generated builds a proprietary dataset that improves model accuracy and become

10.2 One-Line Pitch

"CreditBridge turns a Ghanaian informal worker's MoMo history into a lender-ready credit profile — unlocking loans for the 85% of the workforce that banks currently cannot see."

This opportunity is a hypothesis. Validate it.
Speak to real traders. Sit with a loan officer. Go to Makola or Kejetia. Ask the market woman what happened the last time she tried to get a loan. Then build something worth adopting.